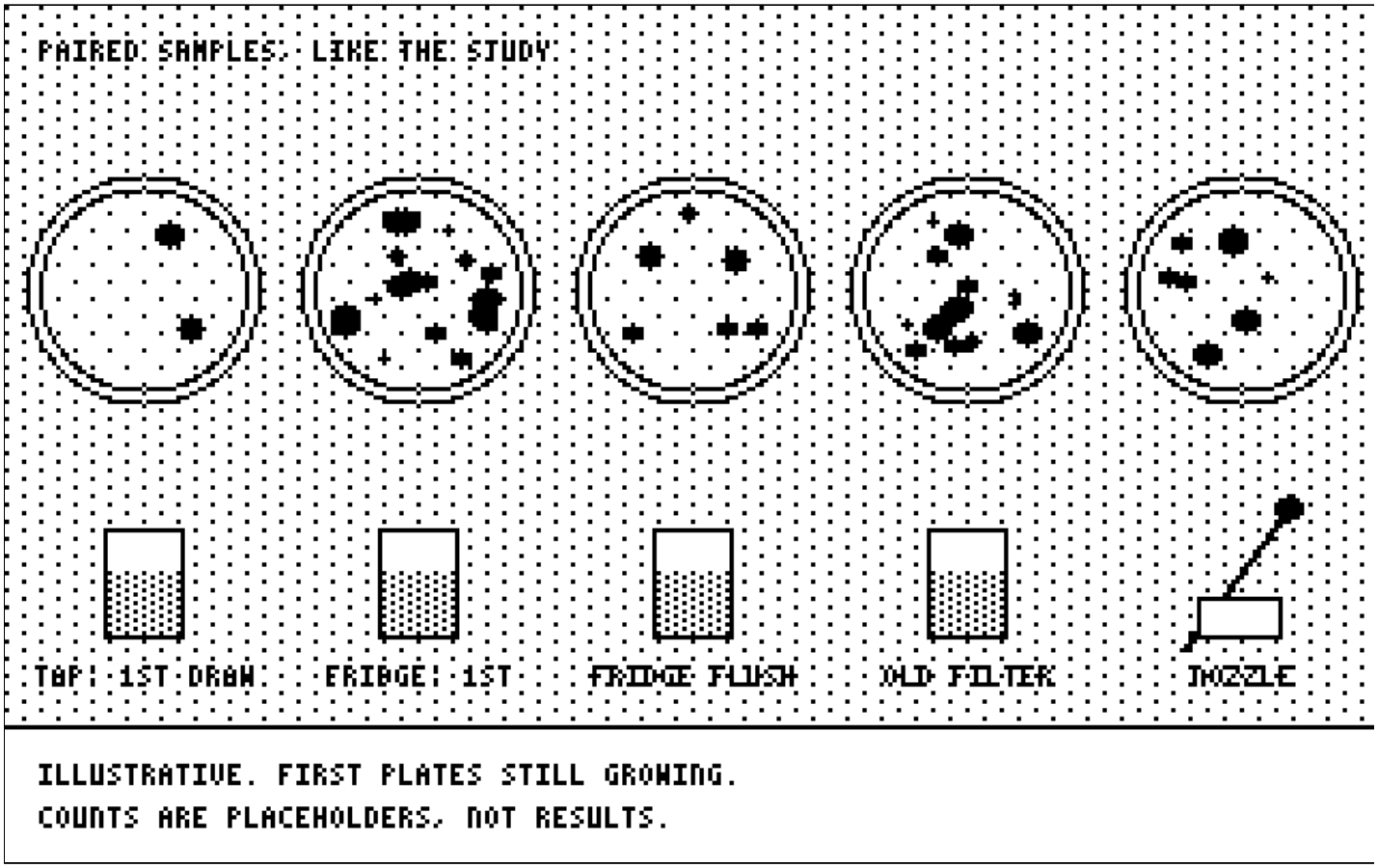
The Hygiene Trap

Scene 22 of 28: The test

Five samples

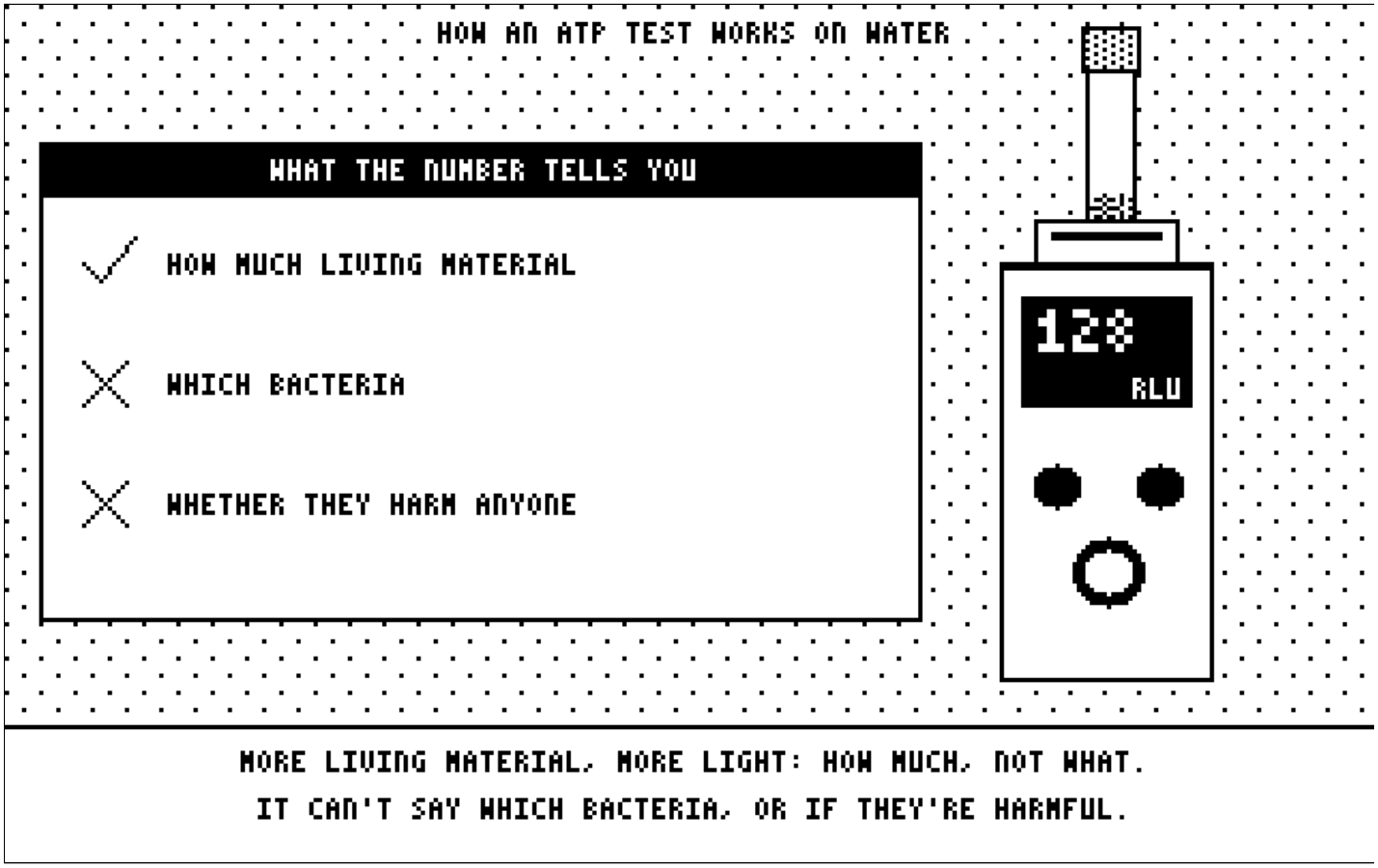


Frame 1. Illustrates line 4: “The first plates are still growing, so there are…”

What the narrator says

1. Here's the part that's more satisfying than speculating. This hypothesis is cheap to test.
2. Take paired samples, as the study did: the fridge's first draw and the tap's first draw, early in the morning, before any water is used. Then a flushed fridge sample, and the nozzle.
3. Test the chlorine in each, and plate the water. If the fridge isn't far above the tap, our hypothesis is in trouble, and that's a useful result.
4. The first plates are still growing, so there are no results here yet.

ATP: how it works

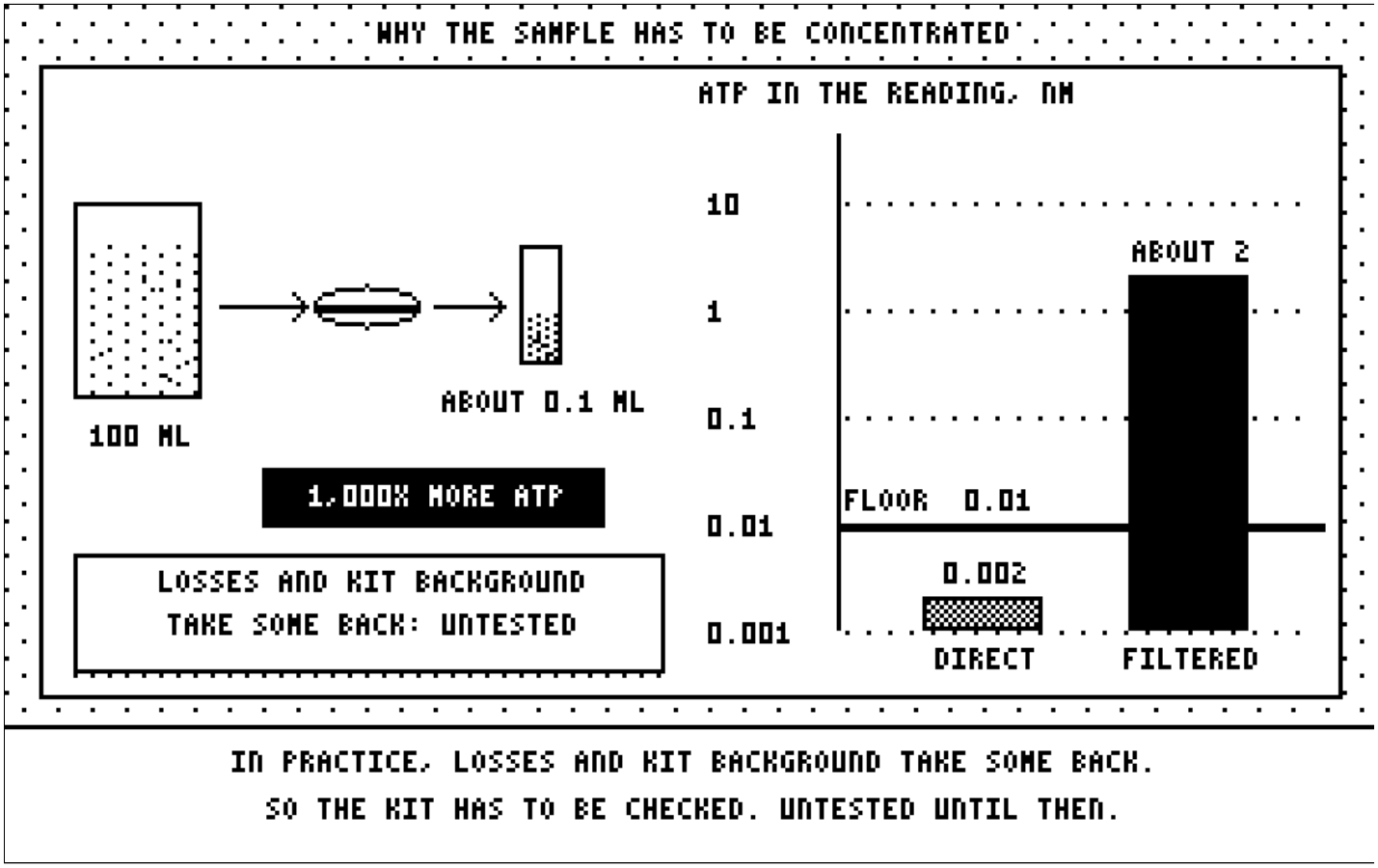


Frame 1. Illustrates line 4: “More living material means more light. It tells you…”

What the narrator says

1. Here's how an ATP test works. Every living cell carries a molecule called ATP, its energy currency.
2. For water, you filter a large sample so the cells are caught on a filter, then a reagent bursts them open and releases their ATP.
3. The ATP meets an enzyme from fireflies, and makes a flash of light. A handheld meter counts the light and shows a number.
4. More living material means more light. It tells you how much, not which bacteria, and not whether they're harmful.

ATP: why concentrate

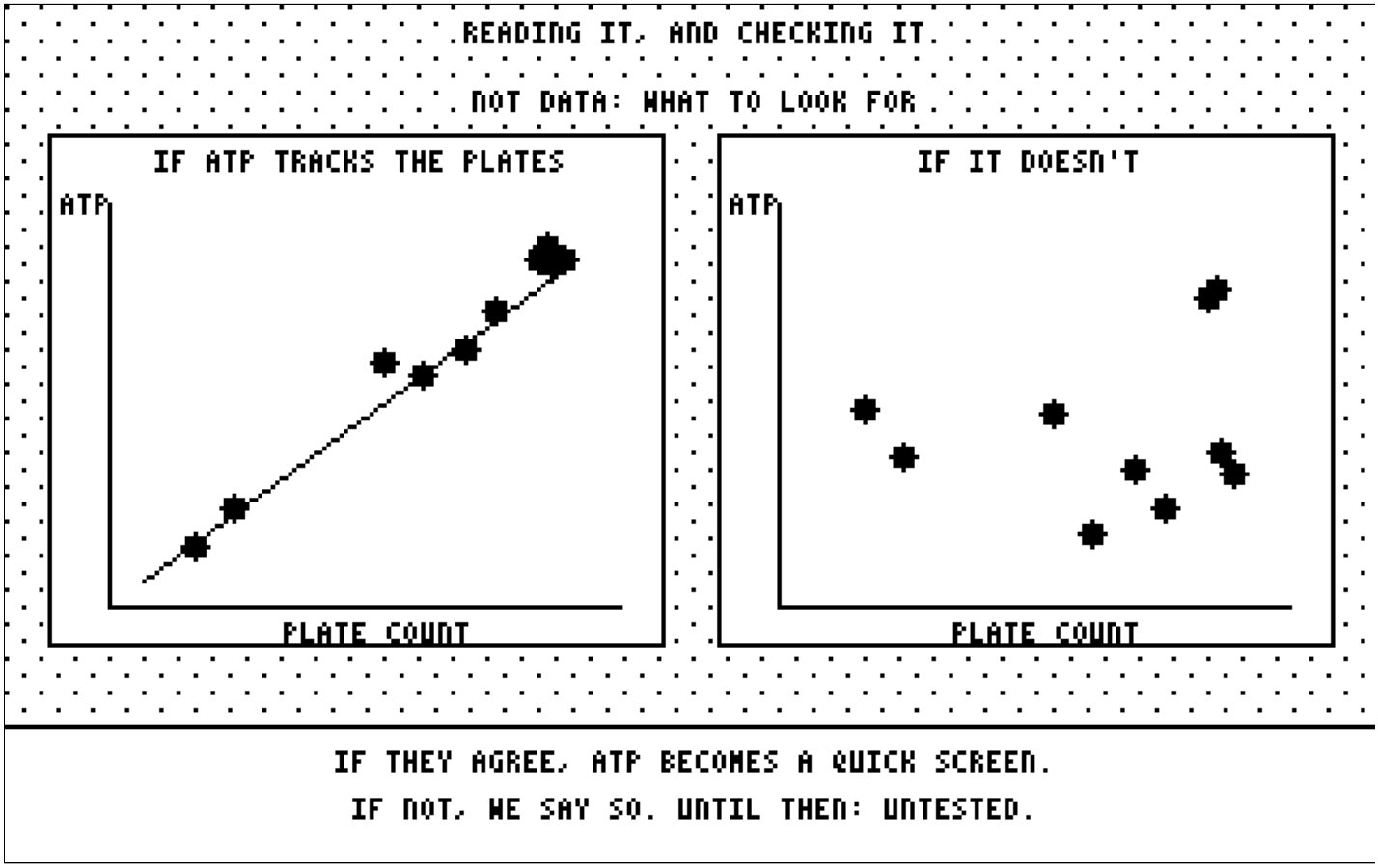


Frame 1. Illustrates line 4: “In practice, losses and kit background take some of…”

What the narrator says

1. The catch is sensitivity. Tap water holds very little ATP, so a handheld meter reading it directly may see nothing.
2. The fix is to concentrate. Filter a hundred millilitres, catch the cells, and burst them into a tiny volume of reagent.
3. By our arithmetic that packs about a thousand times more ATP into the reading, which lifts even tap water above the meter's floor.
4. In practice, losses and kit background take some of that back. So the kit has to be checked, and until it is, it's untested.

ATP: reading and checking it



Frame 1. Illustrates line 4: “If they agree, ATP becomes a quick screen. If...”

What the narrator says

1. When you read it, always run a blank: a sealed bottle. The blank sets the floor.
2. Compare the fridge with the blank first, and with the tap only if the tap rises above the blank.
3. Then the part that matters. Run plates and ATP on the same samples, in many homes, and see whether the two agree.
4. If they agree, ATP becomes a quick screen. If not, we say so. Until then, it's untested.

References and notes

1. Swanson CS, He Q (2024). Linkage of in-refrigerator water dispensing systems to elevated exposure to microbial contamination at the microbiome scale. Am J Infect Control 52:857-859. PMID 38460731. doi:10.1016/j.ajic.2024.03.001
2. Hammes F, Goldschmidt F, Vital M, et al. (2010). Measurement and interpretation of microbial adenosine tri-phosphate (ATP) in aquatic environments. Water Res 44:3915-23. PMID 20605621. doi:10.1016/j.watres.2010.04.015

Notes

- Method: https://github.com/aptitudetechnology/fridge-water/blob/main/replication_protocol.md
- ATP figures in the following pictures are our arithmetic from the average in Hammes 2010 (about 1.75e-10 nmol per cell).

- The 0.01 nM handheld floor is from one study of one meter. Whether concentrating a sample works well enough with a given kit is untested. Method: https://github.com/aptitudetechnology/fridge-water/blob/main/replication_protocol.md
- All readings shown are examples, not data. Validation plan: https://github.com/aptitudetechnology/fridge-water/blob/main/replication_protocol.md

---

Source: The Hygiene Trap, Scene 22 of 25: The test. https://fridge-water.ownhealth.bio/fridge-water#s22 (whole site: https://fridge-water.ownhealth.bio/). Printed November 6, 2026.
Every claim carries a label: established, plausible, untested, anecdote or real. This is a hypothesis, not a diagnosis: for persistent bad breath or symptoms, see a dentist and a doctor.
Pictures and text: CC BY 4.0. Please credit the Fridge water project, https://github.com/aptitudetechnology/fridge-water.